



An Undergraduate Internship on the AssetTrak Web App for Biplobi Barta

By

Salmanur Rahman

Student ID: **2021806**

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Supervisor:

Bijoy Rahman Arif

Lecturer A

Department of Computer Science & Engineering

Independent University, Bangladesh

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Department of Computer Science & Engineering

Independent University, Bangladesh

Attestation

I, Salmanur Rahman, hereby attest that the "AssetTrak Web App" is entirely my own original work, completed with the assistance and direction from my organizational supervisor, MD. Rasel Hossain. It has not been submitted to any other academic institution for the award of any degree or diploma and was created with the information, experiences, and abilities I acquired while working as an intern at Biplobi Barta. I attest that the work presented in this report was completed by me under the supervision of MD. Rasel Hossain. All additional sources of information have been properly recognized.

Signature

Date

Salmanur Rahman

Name

Acknowledgement

I would like to start by sincerely thanking my academic supervisor, **Bijoy Rahman Arif**, who is a lecturer in the Department of Computer Science and Engineering at Independent University, Bangladesh. His constant direction, tolerance, and insightful criticism have been invaluable in the development of this project. I truly appreciate his guidance, which has enabled me to get past the difficulties I had while working on the project, **AssetTrak Web App**.

Additionally, I am grateful to **Biplobi Barta** for giving me the opportunity to work on this project within their organization. I would especially like to thank my supervisor, **MD. Rasel Hossain**, the Head of IT & Multimedia, for his unwavering leadership, support, and direction during the project. Making this project a reality required cooperation with everyone at Biplobi Barta, and I sincerely appreciate their help.

Lastly, I want to express my gratitude to Independent University, Bangladesh's **Department of Computer Science and Engineering** for providing the tools and resources required to complete this project.

Letter of Transmittal

14 January 2026

Bijoy Rahman Arif

Lecturer A

Department of Computer Science and Engineering

School of Engineering, Technology and Sciences

Independent University, Bangladesh

Subject: Submission of Internship Report

Dear Sir,

As a requirement for the completion of my Bachelor program at Independent University, Bangladesh, I am hereby submitting my internship report. The internship was conducted at **Biplobi Barta**, a News Media company where I had the opportunity to develop a web app called '**AssetTrak Web App**'.

This report summarizes the work I performed, the skills I acquired, and the knowledge I applied during my internship period. I am hoping that the report meets your expectations and provides insights into my contributions and learnings from this experience.

I appreciate your support and direction over my academic career, especially during this internship project. Please contact me for any queries regarding this report.

Thank you for your time and consideration.

Yours sincerely,

Salmanur Rahman

ID: 2021806

Evaluation Committee

Supervision Panel

.....	
Academic Supervisor	Industry Supervisor

Panel Members

.....	
Panel Member 1	Panel Member 2
.....	
Panel Member 3	Panel Member 4

Office Use

.....	
Internship Coordinator	Head of the Department
.....	
Industry Coordinator of the Department	

Abstract

In the fast-paced world of modern journalism, the ability to quickly locate and legally verify media assets is the difference between a successful broadcast and a legal liability. For the team at Biplobi Barta, this process was historically defined by a "fragmented chaos" where high-resolution footage and crucial archives were scattered across dozens of physical hard drives, personal USB sticks, and disconnected cloud accounts. This project introduces AssetTrak, a custom-built Digital Asset Management (DAM) system designed to serve as a centralized "Single Source of Truth" for the organization. Created with a modular PHP/MySQL-backend and a responsive design utilizing HTML, Tailwind CSS enhanced by JavaScript for dynamic client-side interactivity, AssetTrak transcends the concept of simple file storage. AssetTrak runs on a highly intricate "one-to-many" relational database configuration, giving journalists the ability to assign different files, such as Raw Edits, Hi-Resolution Images, and Documents, to one master description. The core component of this development effort is the "Safety Net" notification system, which is essentially an automated logic engine checking license expiration dates. By providing visual warnings for aging content, the system effectively removes the human error factor from copyright compliance, offering editors a much-needed "peace of mind." The system's architecture also prioritizes security through a "Gatekeeper" approval model, ensuring that only vetted staff can access or contribute to the repository. Evidence from internal trials and feedback suggests that there has been a paradigm shift in the work environment, where the "deadline panic" of rummaging through hardware has been replaced by a harmonized and searchable digital process. Finally, AssetTrak offers a viable, scalable, and morally sound solution that not only maintains the media history of Biplobi Barta but also allows its producers the creative freedom of storytelling rather than organizing files.

Keywords— AssetTrak, Digital Asset Management (DAM), Media Repository, License Tracking, Web Development, PHP, JavaScript, MySQL, Workflow Optimization, Institutional Archive, Biplobi Barta.

Contents

Attestation	i
Acknowledgement	ii
Letter of Transmittal	iii
Evaluation Committee	iv
Abstract	v
1 Introduction	1
1.1 Overview/Background of the Work	1
1.2 Problem Statement	1
1.3 Objectives	2
1.4 Scopes	2
1.4.1 User Management & Security	2
1.4.2 Asset Ingestion	3
1.4.3 Dashboard & Monitoring	3
1.4.4 Search & Retrieval	3
1.4.5 Access Control	3
2 Literature Review	4
2.1 Relationship with Undergraduate Studies	4
2.2 Related Works	5
3 Project Management & Financing	7
3.1 Work Breakdown Structure (WBS)	7
3.2 Process/Activity wise Time Distribution (Critical Path Method)	8
3.3 Gantt Chart	8
3.4 Process/Activity wise Resource Allocation	9
3.5 Estimated Costing	9
4 Methodology	11
4.1 Requirement Gathering	11
4.2 System Design	12

4.3	Development	12
4.4	Testing	13
4.5	Deployment	13
5	Body of the Project	15
5.1	Work Description	15
5.2	Requirement Analysis	15
5.2.1	Rich Picture	15
5.2.2	Functional and Non-Functional Requirements	17
5.3	System Analysis	17
5.3.1	Six Element Analysis	17
5.3.2	Feasibility Analysis	18
5.3.3	Problem Solution Analysis	18
5.3.4	Effect and Constraints Analysis	18
5.4	System Design	18
5.4.1	Entity Relationship Diagram (ERD)	18
5.4.2	System Architecture	19
5.5	Implementation	19
5.5.1	User Registration Page	20
5.5.2	User Login Page	20
5.5.3	Dashboard Page	21
5.5.4	My Account Page	22
5.5.5	User Approvals Management Page (Admin Only)	24
5.5.6	Upload Activity Page (Staff Only)	24
5.5.7	Asset Approvals Page (Editor Only)	25
5.6	Testing	26
5.6.1	Input and Output Analysis	26
5.6.2	Designing Test Cases	26
5.6.3	Test Results	26
6	Results and Analysis	27
6.1	Functional Outcomes	27
6.2	Performance Evaluation	28
6.3	User Feedback	28
6.4	Impact On Managing Digital Assets and Workflow	29
7	Project as Engineering Problem Analysis	30
7.1	Sustainability of the Project/Work	30
7.2	Social and Environmental Effects and Analysis	31
7.3	Addressing Ethics and Ethical Issues	31

8 Lesson Learned	32
8.1 Problems Faced During this Period	32
8.2 Solution of those Problems	33
9 Code	34
9.1 Source Code	34
10 Sustainability, Future Work & Conclusion	40
10.1 Sustainability along with Environmental and Social Effects	40
10.2 Future Works	41
10.3 Conclusion	41
Bibliography	44

List of Figures

3.1	Work Breakdown Structure (WBS) of AssetTrak	7
3.3	Gantt Chart Schedule for AssetTrak	8
5.2.1	Rich Picture	16
5.4.1	Entity Relationship Diagram	18
5.4.2	System Architecture	19
5.5.1	User Registration Page	20
5.5.2	User Login Page	20
5.5.3.	Dashboard Page for Admin	21
5.5.3.	Dashboard Page for Staff	22
5.5.3.	Dashboard Page for Editor	23
5.5.4	My Account Page	23
5.5.5	User Approvals Management Page	24
5.5.6	Upload Activity Page	24
5.5.7	Asset Approvals Page	25
9.1.1	db.php	34
9.1.2	functions.php	35
9.1.3	sidebar.php	35
9.1.4	login.php	36
9.1.5	register.php	36
9.1.6	dashboard.php	37
9.1.7	user management.php	37
9.1.8	my account.php	37
9.1.9	get files.php	38
9.1.10	upload_activity.php	38
9.1.11	asset_approval.php	39
9.1.12	logout.php	39
10.3.	Internship Offer Letter	42
10.3.	Letter of Completion	43

List of Tables

3.2	Critical Path Analysis	8
3.4	Resource Allocation	9
3.5	Estimated Project Costing	10

Chapter 1

Introduction

1.1 Overview/Background of the Work

In the fast-moving news media environment, the method of quickly accessing, verifying, and using digital assets such as images, video footage, audio clips, and documents becomes very crucial for everyday production. In my internship at Biplobi Barta, a startup news media company based in Bangladesh, managing these ever-growing libraries of content is a major operational challenge that I have seen.

The project, titled “AssetTrak,” is a direct answer to the organization’s needs. It is a web-based DAM system for serving as a secure, centralized repository for the media files of the company. Unlike off-the-shelf commercial options available in the market, such as DAAM or MAAM software, which are too expensive or over-engineered for a startup, AssetTrak is designed to exactly fit the scale and workflow of Biplobi Barta.

This project was developed under the supervision of my organizational supervisor, Mr. MD Russel Hossain. In a few discussions with him regarding the bottlenecks in the workflow of the current asset management, we decided that a custom-made web application would be the most appropriate way to shift the company from scattered physical storage to a streamlined digital cloud-based system.

1.2 Problem Statement

Before AssetTrak was developed, Biplobi Barta had a digital storage system that was decentralized and manual. The workflow relied very heavily on local hard drives and disparate personal cloud storage accounts. This led to several critical issues:

- **Inaccessibility and Inefficiency:** Journalists and editors couldn’t find particular files quickly. Often, assets were buried deep inside unnamed local drive folders; retrieval was very slow, which delayed the pace at which news would be published.
- **Security Risks:** Without a centralized authentication system, sensitive media assets were at risk from unauthorized access or accidental deletions. There was no role-based

hierarchy to regulate who can modify and delete data.

- **Legal and Compliance Risks:** One major oversight of the old system is the inability to track license expiry dates for purchased or licensed media. Using an image or video with an expired license opens the company to legal risks.
- **Lack of Remote Collaboration:** Reliance on physical hard drives meant that staff had to be physically present at a specific workstation to access files, something not efficient for a modern, mobile news team.

1.3 Objectives

The main goal of this project was to design and implement a secure, web-based system for the digitization and centralization of the asset management of Biplobi Barta.

Specific objectives included:

- **Centralization:** To establish a single source of truth where all digital assets (images, PDFs, video, audio) are kept in one place and easily accessible via a web browser.
- **Role-Based Security:** The authentication system should be implemented to support different classes of users such as Admin and Staff. Only the administrators should be authorized to approve new user accounts or delete sensitive assets.
- **Risk Mitigation:** Integrating an automated tracking system to notify users when asset licenses have expired or are set to expire within the next 30 days to avoid copyright infringement.
- **Efficiency:** To offer strong search functionality to the users, enabling them to find any asset in an instance using tags, titles, or descriptions, and batch uploading for accelerating news material ingestion.

1.4 Scopes

The AssetTrak system addresses the entire digital asset management lifecycle, from user registration to asset archival. The functional boundaries of the AssetTrak system include:

1.4.1 User Management & Security

- A secured Login and Sign-up interface that takes minimal credentials: email, password, and full name.
- Having a strict Admin Approval Queue keeps newly registered users in a “pending” state until an Admin explicitly validates and approves their account.
- Role management capabilities allowing Admins to promote Staff members or revoke access.

1.4.2 Asset Ingestion

- Support for multiple file formats relevant to news media, including PDF, text, images, videos, and audio.
- A Batch Upload feature to let users upload their assets with associated metadata such as Batch Title, Descriptions, and Tags for better organization.

1.4.3 Dashboard & Monitoring

- A dynamic dashboard that displays, in real time, the “License Expiry Alert” panel at the top, showing the assets that have expired or will be expiring soon—for example, within 2 days or 28 days ago.
- Visual indicators of status for assets, including “Active”, “Expiring Soon”, and “Expired” to drive decision-making.

1.4.4 Search & Retrieval

- A working search bar that allows users to query the database by asset title, description, or tags, with the functionality to clear applied filters instantly.

1.4.5 Access Control

- While all users can view and download assets, the Delete function is strictly restricted to Admin users to prevent data loss.

Chapter 2

Literature Review

2.1 Relationship with Undergraduate Studies

In developing **AssetTrak**, the project was not strictly a coding exercise; rather, it applied theoretical and technical underpinnings that were established in my undergraduate studies. Conceptual elements from several key courses in my curriculum for Computer Science were directly applied to the project:

- **Database Management Systems (DBMS):** This class was essential to the design of the back end. Applying relational database concepts, I designed the MySQL schema by creating normalized tables for **users**—storing credentials and roles—and **assets**, which stored metadata such as batch titles, tags, and expiry dates. The project involved writing complex SQL queries that would handle data insertion, retrieval, and updates efficiently.
- **Web Application and Internet:** The core development of the application relies heavily on this course. I used PHP for the server-side logic, managing sessions for secure login, and handling file uploads. In regards to the frontend implementation using HTML, Tailwind CSS, and JavaScript for a responsive UI and dynamic alerts—for example, the `confirmDeletion` script—the web technologies learned in class were in direct application.
- **System Design Analysis:** This course helped in defining the system architecture and user roles. I defined two different classes of users—**Admin** and **Staff**—and mapped their permissions; for example, only Admins can delete assets or approve users. The analysis phase involved creating a workflow that will mimic real-world operations of a news agency, ensuring the system is scalable and secure.
- **Software Engineering & Software Project Management:** These courses gave me the whole picture of SDLC. From gathering requirements from my supervisor, Mr. MD Russel Hossain, to the implementation of a “License Expiry Alert” feature mitigating legal risks, I have been able to follow structured problem-solving. It also guided the testing phase to ensure the system effectively handles different file types (PDF, video, audio) without errors.

- **Data Structures and Algorithms:** In a media environment, efficient retrieval of data is paramount. Search and sorting algorithms were applied to implement the functionality of the search bar so that users could filter the assets by tags, titles, or descriptions in real time. This ensures that even with the growth of the database, retrieval time remains low.

2.2 Related Works

The concept of digital asset storage in news media falls under two major industry categories: **Digital Asset Management (DAM)** and **Media Asset Management (MAM)**. While DAM systems are broad repositories for all file types—images, documents, branding—MAM systems are specifically tailored for broadcasting workflows, focusing on video and audio time-based assets.

The requirement for such tools in the media industry has been catered to by several systems and research papers already:

1. Enterprise DAM/MAM Solutions

Large news agencies tend to use commercial applications such as **Avid MediaCentral** or **Dalet Galaxy** for this purpose. These are powerful MAM solutions applied in high-end broadcast automation. However, they are often too complex and expensive for startups like *Biplobi Barta*. There are also free and open-source alternatives, such as **ResourceSpace**, which offer web-based access and metadata tagging similar to AssetTrak but often take significant resources on the server to maintain.

2. Academic Research & Similar Systems

Research in the field emphasizes the need for shifting from manual storage to centralized systems.

- **TVRI Broadcasting Case Study:** Dhany and Bandung (2016) have proposed a DAM architecture for a public broadcasting organization. Their work highlighted that with manual asset management, there is a great loss of data and a hindrance to efficiency. Similar to AssetTrak, their system also concentrated on metadata management to enhance the searchability in a fast news environment [1].
- **Web-Based Asset Management:** Setiadi and Rangkuti (2022) developed a web-based asset management system emphasizing the usage of modern web frameworks to handle asset maintenance and tracking. Their findings corroborate the need for automated tracking—such as the “Expiry Date” feature in AssetTrak—to optimize the lifespan and legal compliance of digital assets [2].
- **Centralized Information Systems:** Dewi and Ririd (2019) addressed the inefficiencies of manual asset recording by designing a web-based information system. Their research

emphasized that a centralized database significantly reduces data redundancy and the risk of asset loss. This aligns with AssetTrak's core objective of serving as a "Single Source of Truth," ensuring that all organizational media is stored in and retrieved from a unified, secure repository [3].

- **System Implementation and Accessibility:** In the study carried out by Anggoro and Hidayat (2020) on the design and implementation of asset systems for the purpose of providing accessibility for remote access and monitoring in real time, the significance of user-friendly interfaces for the use of the system has been underscored. This supports AssetTrak's architectural choice of using a responsive design (Tailwind CSS) and Role-Based Access Control (RBAC), allowing Admins, Editors, and Staff to interact with the system securely and efficiently from any location [4].

AssetTrak is a lightweight, purpose-built application distinguished from these generic or high-cost solutions. It combines core storage features of a DAM with specific "License Expiry" and "Batch Upload" needs of *Biplobi Barta* and hence provides them a cost-effective alternative to the heavy "DAAM and MAAM" software which larger agencies typically use.

Chapter 3

Project Management & Financing

Effective project management means that software development meets its deadlines, stays within budget, and fulfills the requirements of the user. In managing **AssetTrak**, I applied standard project management techniques, including WBS, CPM, and Gantt Charts, which track progress from November 7, 2025, the project start date, up to December 25, 2025, the completion date.

3.1 Work Breakdown Structure (WBS)

The WBS is a hierarchical breakdown of the entire scope of work that the project team must complete. AssetTrak was divided into five major phases aligned with the SDLC: Requirement Analysis, System Design, Development, Testing, and Deployment. Each phase was then divided into specific, manageable tasks, that is, work packages. This granular approach helped ensure that no critical feature, such as the “License Expiry Alert” or “Batch Upload,” would get missed out on in this short window for development.

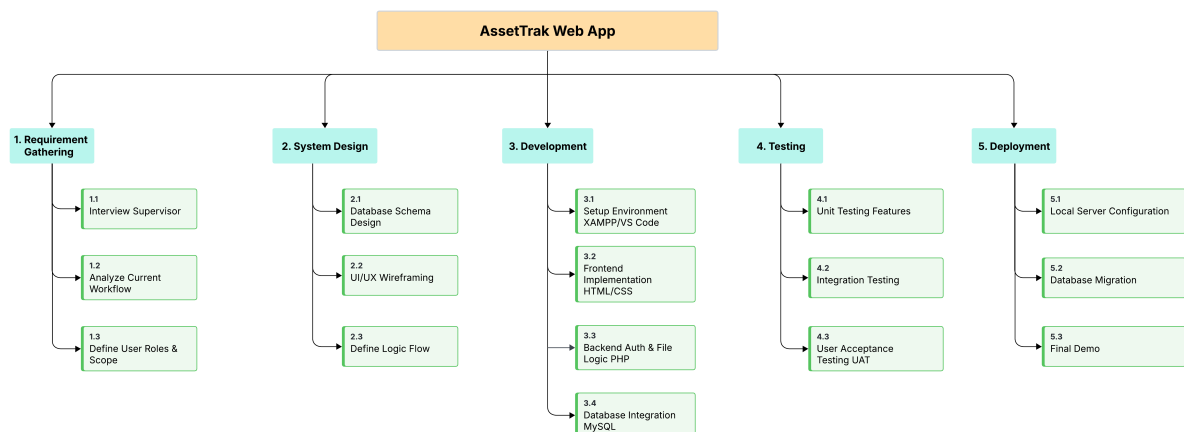


Figure 3.1: Work Breakdown Structure (WBS) of AssetTrak

3.2 Process/Activity wise Time Distribution (Critical Path Method)

The Critical Path Method was applied to trace the longest sequence of dependent tasks that determine the shortest possible duration of the project. Since the internship timeline was fixed, identification of the critical path was essential in prioritizing backend logic and database integration over non-essential polishing of UI.

The table below shows the activities and their duration and dependencies.

ID	Activity Name	Duration	Predecessors	Start Date	End Date
A	Requirement Gathering	5 Days	-	07 Nov	11 Nov
B	System Design (DB & UI)	6 Days	A	12 Nov	17 Nov
C	Environment Setup	1 Day	B	18 Nov	18 Nov
D	Backend Dev (Auth & Logic)	12 Days	C	19 Nov	30 Nov
E	Frontend Dev (UI)	8 Days	C	19 Nov	26 Nov
F	Integration (PHP + MySQL)	5 Days	D, E	01 Dec	05 Dec
G	Testing (Unit & UAT)	10 Days	F	06 Dec	15 Dec
H	Bug Fixing & Refinement	6 Days	G	16 Dec	21 Dec
I	Deployment & Documentation	4 Days	H	22 Dec	25 Dec

Table 3.2: Critical Path Analysis

Critical Path: A → B → C → D → F → G → H → I

Note: Task E (Frontend) runs parallel to Task D (Backend) but Task D takes longer, making it the critical constraint.

3.3 Gantt Chart

A Gantt Chart was used to visually track the project schedule. It shows the beginning and ending dates of the project’s summary and terminal components. This visualization helped in monitoring progress against the deadline of December 25, ensuring that the “Testing” phase had sufficient time allocated before the final handover.

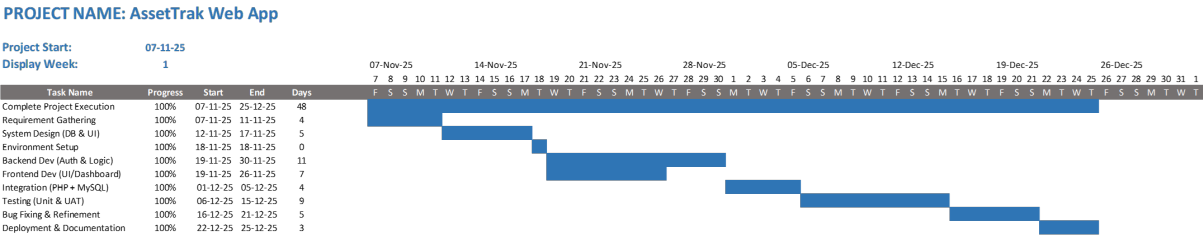


Figure 3.3: Gantt Chart Schedule for AssetTrak

3.4 Process/Activity wise Resource Allocation

Resource allocation involves assigning the available resources (human, hardware, and software) to specific tasks to maximize efficiency. Since this was an individual internship project, the primary human resource was myself (Developer/Intern), supervised by Mr. MD Russel Hossain. The table below details the tools and effort allocated to each phase.

Activity Phase	Human Resource	Software/Tools Resources	Re-	Hardware	Output
Requirement Gathering	Developer, Supervisor	MS Word (Docs), Google Meet		Laptop	SRS Document
System Design	Developer	Draw.io, Figma (Wire-frame)		Laptop	Schema, UI
Development	Developer	VS Code, XAMPP (Apache, MySQL)		Laptop	Codebase
Testing	Dev, Supervisor	Chrome DevTools, PHPUnit		Laptop	Bug Report
Deployment	Developer	XAMPP Control Panel, Windows OS		Laptop	Deployed App

Table 3.4: Resource Allocation

3.5 Estimated Costing

Since this was an academic internship project for a startup, the direct monetary costs of developing it were kept to a minimum using open-source technologies: PHP, MySQL, and Apache. In a professional estimation, though, the “Cost” signifies the value of time, electricity, and internet data that might have gone into it, or any software licensing if this were a commercial contract. Values are estimated in Bangladeshi Taka.

3.5. ESTIMATED COSTING

SDLC Phase/Activity	% Effort	Cost Description	Cost (BDT)
Requirement Analysis	10%	Communication & Documentation costs	2,000
System Design	15%	Design tools & Planning man-hours	3,500
Development	40%	Coding man-hours (approx. 100 hrs)	10,000
Testing & Debugging	20%	Quality Assurance effort	5,000
Deployment & Training	10%	Implementation & Supervisor Training	2,500
Miscellaneous	5%	Internet, Electricity, & Contingency	2,000
Total Estimated Cost	100%	Total Project Value	25,000

Table 3.5: Estimated Project Costing

Chapter 4

Methodology

This chapter describes the structured process followed for developing **AssetTrak**. The project was developed based on the **System Development Life Cycle (SDLC)** methodology, employing an iterative development model. This allowed for constant feedback to be received from the organizational supervisor, Mr. MD Russel Hossain, about how the software should evolve to fulfill the specific operational needs of Biplobi Barta. The different phases of this development process include Requirement Gathering, System Design, Development, Testing, and Deployment.

4.1 Requirement Gathering

The foundational requirement gathering was done to identify the specific problems related to the storage of digital assets at Biplobi Barta, and in doing so, it facilitated direct consultation and interviews with the organizational supervisor and editorial staff to understand their day-to-day workflow bottlenecks. These discussions revealed that the current setup of relying on local hard drives and personal cloud accounts was unsustainable. The main needs identified were:

- **Centralization:** The need for a single, web-based repository accessible through a browser, without having to physically transfer the files themselves.
- **Security & Access Control:** The system needed a strict difference in privileges between **Admin** and **Staff**. It was decided that only Admins should have high-level privileges such as the approval of new user registrations and the deletion of assets to prevent unauthorized loss of data.
- **Legal Compliance:** The most important feature to implement was the “License Expiry Alert.” The system had to calculate the days remaining on an asset’s license automatically and highlight this to the user whenever an asset was due to expire within 30 days.
- **Searchability:** The ability to filter assets with tags, titles, or descriptions was considered a must in order to reduce search time by journalists for certain archival footage.

4.2 System Design

The system architecture was designed to be robust, scalable, and user-friendly based on the gathered requirements.

- **Database Schema Design:** Using **phpMyAdmin**, a schema for a relational database was designed in MySQL. Two main tables were normalized:
 - **Users Table:** It stores user credentials and hashed passwords, along with role identifiers for Admin/Editor/Staff. The ‘status’ field was for handling Pending/Approved logic at the time of new registrations.
 - **Assets Table:** Designed to store metadata such as `batch_title`, `description`, `file_type`, `tags`, and most importantly the `license_expiry_date`.
- **User Interface Design:** The UI was wireframed with simplicity in mind. A dashboard layout was chosen where the most critical information, the “Expiry Alerts,” is shown right at the top for immediate visibility. Navigation was sidebar-based for Admins, showing “User Approval,” and streamlined for Staff.
- **Logic Flow:** An exercise in systems logic was done to ensure data integrity. For example, the “Delete” function was designed to first prompt a double confirmation (`confirmDeletion` script) to avoid accidental deletions—a necessary safety feature in a newsroom environment.

4.3 Development

In the development phase, translation of the design into a functional application required applying certain software and development tools.

- **Development Environment:** For the development of this project, the main IDE used was **Visual Studio Code (VS Code)**. A local server environment was created using **XAMPP**, which provided the **Apache** web server to handle HTTP requests and the **MySQL** database server for storing data.
- **Frontend Implementation:** The client-side interface used **HTML5** for structure and **Tailwind CSS** for styling. The design was made fully responsive to ensure that journalists would have the ability to access the system from laptops or tablets. **JavaScript** was used in dynamic client-side interactions, such as confirmation dialogs during asset deletions and real-time form validation during the sign-up process.
- **Backend Implementation:** **PHP** was chosen for server-side scripting because it was compatible with the server infrastructure already in place. The following backend features were designed and integrated:

- **Session Management:** Using `session_start()` to maintain secure user login states.
- **File Handling:** Wrote logic to validate MIME types (only valid images, videos, and PDFs can be uploaded) and securely move files to the `uploads/` directory.
- **Date Arithmetic:** Special PHP code comparing the current server date against the asset's `license_expiry_date` has been developed and included in the generation of the dashboard alerts.

4.4 Testing

Rigorous testing was done to ensure that the system was bug-free and met the initial objectives. Testing included:

- **Unit Testing:** Individual components, such as the login authentication function and the file upload mechanism, were independently tested. For instance, the system was tested to ensure that users in “Pending” status were rejected even when the password was correct.
- **Integration Testing:** This included making sure the PHP backend communicated accurately with the MySQL database. I checked that, on uploading a batch file by a user, the file was correctly stored in the local folder while the metadata was duly inserted into the database table.
- **UAT:** The system was demonstrated to the supervisor, Mr. MD Russel Hossain. We gave a mock demonstration, including scenarios of a Staff member trying to delete a file, which was blocked successfully, and an Admin approving a new user. For testing the “Expiry Alert,” we inserted manually some assets with dates in the past to see that the red alert banners show up appropriately on the dashboard.

4.5 Deployment

For the purpose of this internship project, the application was deployed in a local environment.

- **Local Hosting:** The web application was locally hosted by configuring the **Apache** server within the XAMPP stack. The project files were deployed to the `htdocs` directory, making the application accessible via `localhost` on the development machine.
- **Database Migration:** The SQL structure was exported and then imported into the host's **phpMyAdmin** interface to set up the live database.
- **Configuration:** Host, User, and Password were set to configure the database connection within these PHP scripts for smooth connectivity.

- **Permissions:** The `uploads/` directory was granted write permissions by explicit declaration to ensure that the server could save media uploaded by the user without permission errors.

The proof-of-concept was successfully demonstrated with this deployment setup, which proved that **AssetTrak** is ready to be migrated to a live cloud server in the future for use organization-wide.

Chapter 5

Body of the Project

This chapter details the systematic approach taken to develop **AssetTrak**, covering the transition from initial requirements to a fully modular, industry-standard web application.

5.1 Work Description

The primary objective was to build a **centralized media repository** for Biplobi Barta to replace unorganized local storage. The work involved:

- **Database Architecting:** Creating a schema for a relational database to support the data structures necessary to describe the **one-to-many relationships** between the meta-data of assets and multiple files.
- **Modular Backend Development:** Implementing the logic in **PHP**, adhering to professional standards where the different configurations are separated into different PHP files such as `db.php` for configuring the databases and `functions.php` for helper functions.
- **Dynamic UI/UX: Tailwind CSS** enabled the designs of a responsive, one-page experience on a multi-page structure even though the app itself is multiple pages.
- **Security Implementation:** Developing a **secure authentication system** featuring password hashing and a dual-layer **"Admin-Approval" gate** for new users.

5.2 Requirement Analysis

5.2.1 Rich Picture

The **Rich Picture** is more than an illustrated flowchart; it is instead a tool of visual storytelling that seeks to represent "messy" human and organizational realities particular to a project. In relation to **AssetTrak**, for example, The Rich Picture depicts an improvement from a disorganized and stressful media environment that existed in Biplobi Barta.

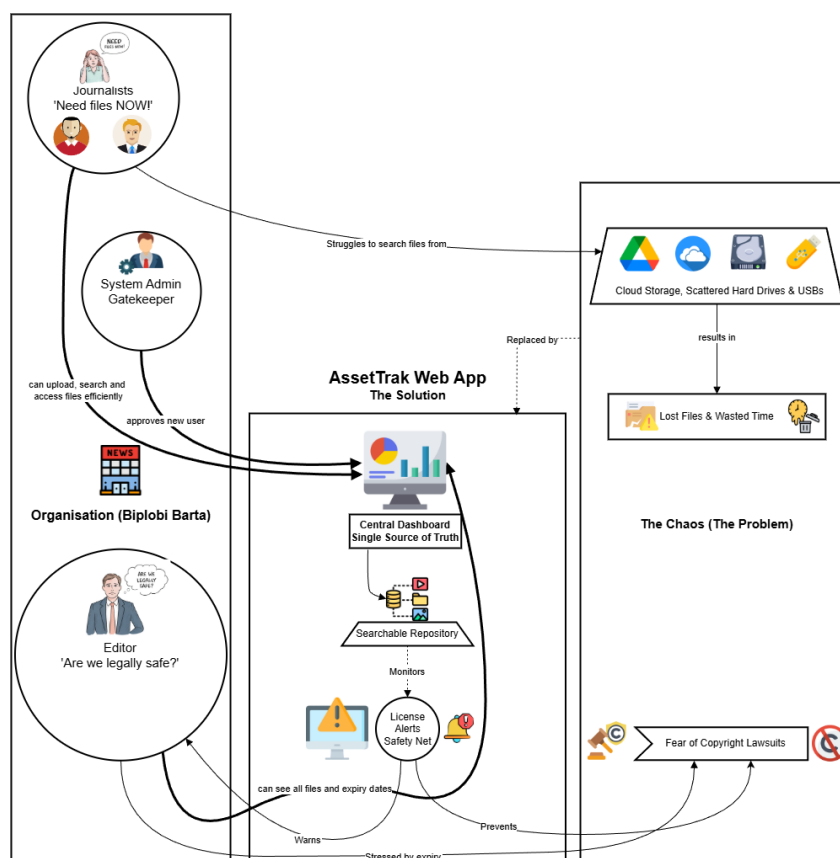


Figure 5.2.1: Rich Picture

Description: The Rich Picture above serves as a snapshot of the operational transformation facilitated by this project. It is divided into two contrasting worlds: the **"Old Way"** of chaos and the **"New Way"** of structured management.

- The Problem Context (The "Old Way"):** Initially, the media management at Biplobi Barta was plagued by **decentralization**. Journalists were often seen in a state of high-pressure urgency, physically digging through scattered hard drives and USB sticks to find specific coverage files. This **"fragmented storage"** led to significant time wastage and a lack of accountability. More critically, the diagram highlights the **"Legal Stress"** faced by Editors. Without this central database, there was always this lurking fear of using media with licenses that were no longer valid, which presented a real risk of copyright infringement lawsuits.
- The Key Stakeholders:** The role of human agents is positioned at the centre of the issue. The **Journalist** represents speed and easy batch access, the **Editor** represents compliance and safety, and the **System Admin** is the **gatekeeper** ensuring that only authorized individuals enter the repository.
- The Solution Environment (AssetTrak):** By introducing the **AssetTrak Hub**, messy relationships are replaced by a **"Single Source of Truth"**. The **"Safety Net"**—the automated alert engine that monitors license dates—is a vital component. The system

proactively "pushes" alerts, transforming organizational anxiety into "**Peace of Mind**".

- **The Transformation Logic:** Thick flow arrows indicate a transition in behavior. Instead of struggling with hardware, Journalists interact with a **searchable dashboard**. The Admin no longer manages physical drives but manages **digital roles and permissions**.

5.2.2 Functional and Non-Functional Requirements

Functional Requirements:

- **User Management:** Users can register by themselves and Admins can approve or reject registrations.
- **Batch Upload:** Enables the capability to upload files with the same asset title.
- **License Tracking:** Automated alerts will be displayed on the Dashboard for assets expiring within **30 days**.
- **Global Search:** Searching assets via titles or specific metadata tags.

Non-Functional Requirements:

- **Security:** High-level encryption has been used for passwords using `password_hash()`.
- **Usability:** A clean, **sidebar-driven navigation system** for ease of use.
- **Scalability:** **Modular coding** that can be expanded with other functionalities, such as categories, without affecting basic processes.

5.3 System Analysis

5.3.1 Six Element Analysis

1. **Functionality:** The system can perform all functions of the original asset from creation to deletion.
2. **Usability:** Contains easy-to-use interfaces and **color-coded status indicators** (Green: OWNED, Yellow: EXPIRING SOON, Red: EXPIRED).
3. **Scalability:** Modular architecture makes it simple to extend this solution for usage in mobile APIs or cloud-based storage.
4. **Integration:** The application is integrated with the **XAMPP stack** for compatibility with standard web servers.
5. **Performance:** Well-constructed SQL queries using `GROUP_CONCAT` enable quick loading regardless of the file count.
6. **Security:** **Session-based access control** redirects unauthorized users to login.

5.3.2 Feasibility Analysis

- **Technical Feasibility:** Employs **PHP 8.x** and **MySQL**, which are strong and fully supported for this purpose.
- **Operational Feasibility:** The flow resembles other cloud storage services, requiring minimal training.
- **Economic Feasibility:** Entirely **open-source software** used to save on licensing costs.
- **Schedule Feasibility:** The module-based development fits within the **12-week internship** schedule.

5.3.3 Problem Solution Analysis

Previously, media assets were scattered across different computers, leading to license violations and lost files. **AssetTrak** solves this by creating a **single source of truth**. By introducing the "**License Expiry Alert**," the organization moves to a proactive management style.

5.3.4 Effect and Constraints Analysis

The main constraint is the reliance on a **local server environment (XAMPP)**. While effective for internal use, transition to a live environment would require configuring `php.ini` for larger file upload limits.

5.4 System Design

5.4.1 Entity Relationship Diagram (ERD)

The database is the backbone of the AssetTrak system. The ERD below represents a normalized relational structure designed to handle complex "one-to-many" relationships for batch uploads and user tracking.

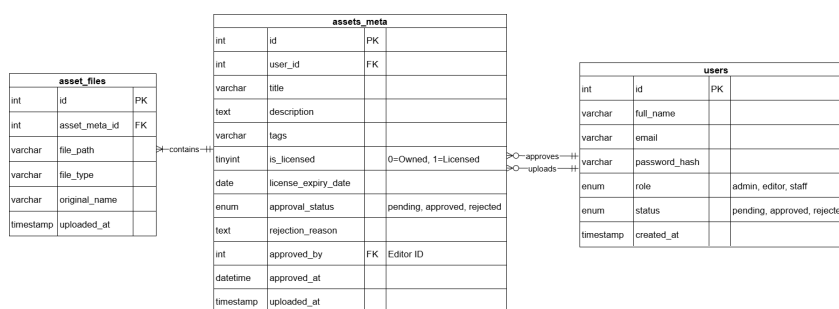


Figure 5.4.1: Entity Relationship Diagram

Detailed Entity Descriptions:

- **Users Entity:** Manages the application gateway, storing credentials, hashed passwords, and the **”Gatekeeper”** logic (approval/role details).
- **Assets_Meta Entity (The Parent):** The central repository for metadata—title, description, tags, and license expiration date. Metadata is **decoupled** from file content, ensuring a unitary asset record even for multiple files.
- **Asset_Files Entity (The Child):** Specifically designed to support **Batch Uploads** through a Foreign Key relationship with Assets.Meta. Every row represents a physical file in the `/uploads/` directory.

5.4.2 System Architecture

AssetTrak follows a **Modular Client-Server Model** for professionalism, maintainability, and debugging.

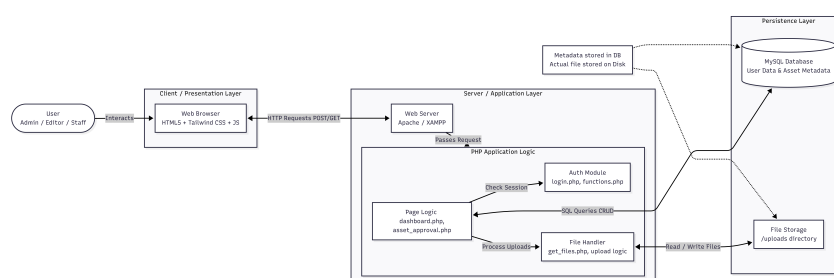


Figure 5.4.2: System Architecture

1. **The Presentation Layer (UI):** Built with Tailwind CSS and served through files like `dashboard.php`, `login.php`, and `my_account.php`. `sidebar.php` unifies navigation and logic for a **seamless transition**.
2. **The Application Logic Layer (PHP):** The system **”brain”**. It uses `functions.php` for global security checks and **modular page files** for specific POST requests. PHP Sessions monitor user identity.
3. **The Data Persistence Layer:** Consists of the MySQL Database and physical File Directory, bridged by `db.php`. Files are **renamed and relocated** to the `/uploads/` folder to avoid naming collisions.

5.5 Implementation

AssetTrak is a modular web application where each file serves a distinct logical purpose.

5.5.1 User Registration Page

The entry point for new staff, collecting details and **hashing passwords**. Accounts are created with a **"pending"** status. A clean form with fields for Name, Email, and Password.

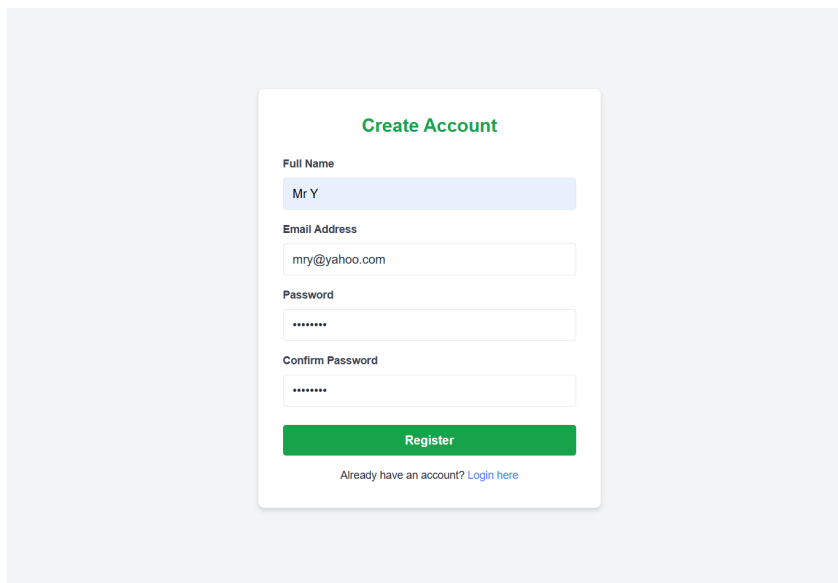
A screenshot of a user registration form titled "Create Account" in green. The form is centered on a light gray background. It contains the following fields: "Full Name" with a dropdown menu showing "Mr Y", "Email Address" with the text "mry@yahoo.com", "Password" with masked characters "*****", and "Confirm Password" with masked characters "*****". A green "Register" button is at the bottom, followed by a link "Already have an account? Login here" in blue.

Figure 5.5.1 : User Registration Page

5.5.2 User Login Page

The default landing page that validates credentials and checks for **Admin approval** before granting a session. A blue-themed login card with error handling alerts).

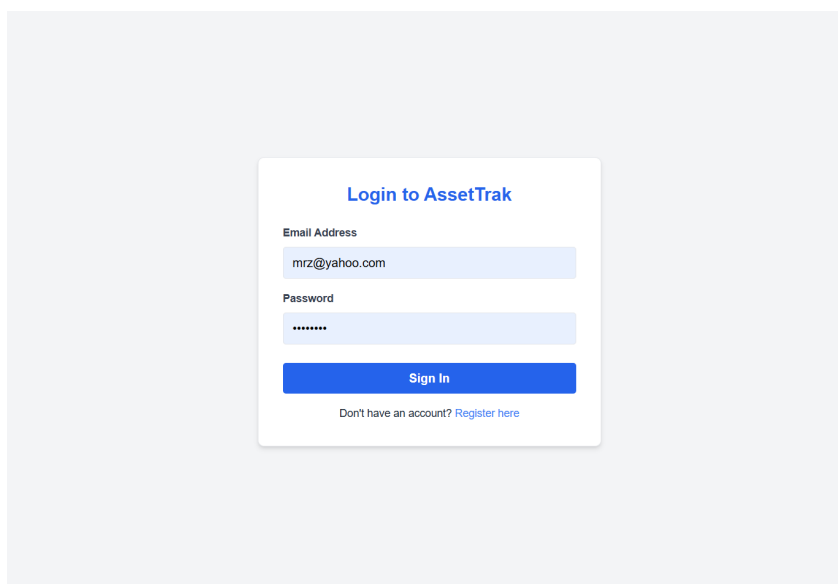
A screenshot of a user login form titled "Login to AssetTrak" in blue. The form is centered on a light gray background. It contains the following fields: "Email Address" with the text "mrz@yahoo.com" and "Password" with masked characters "*****". A blue "Sign In" button is at the bottom, followed by a link "Don't have an account? Register here" in blue.

Figure 5.5.2 : User Login Page

5.5.3 Dashboard Page

The heart of the app. It displays the asset repository, the upload form, and **critical alert banners** for expired or expiring licenses. In Asset Repository section, it shows a dense table view with file preview, title and tags, file description, status badges, expiry date with edit expiry date button, uploaded user, file view, download and delete buttons).

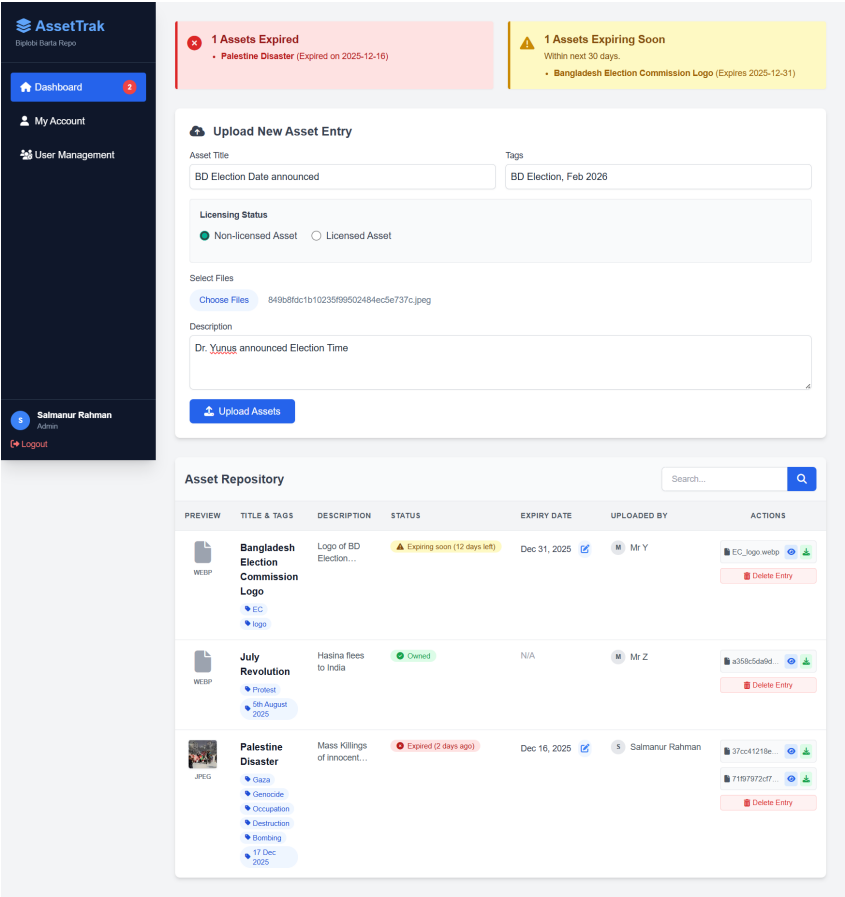


Figure 5.5.3.1 : Dashboard Page for Admin

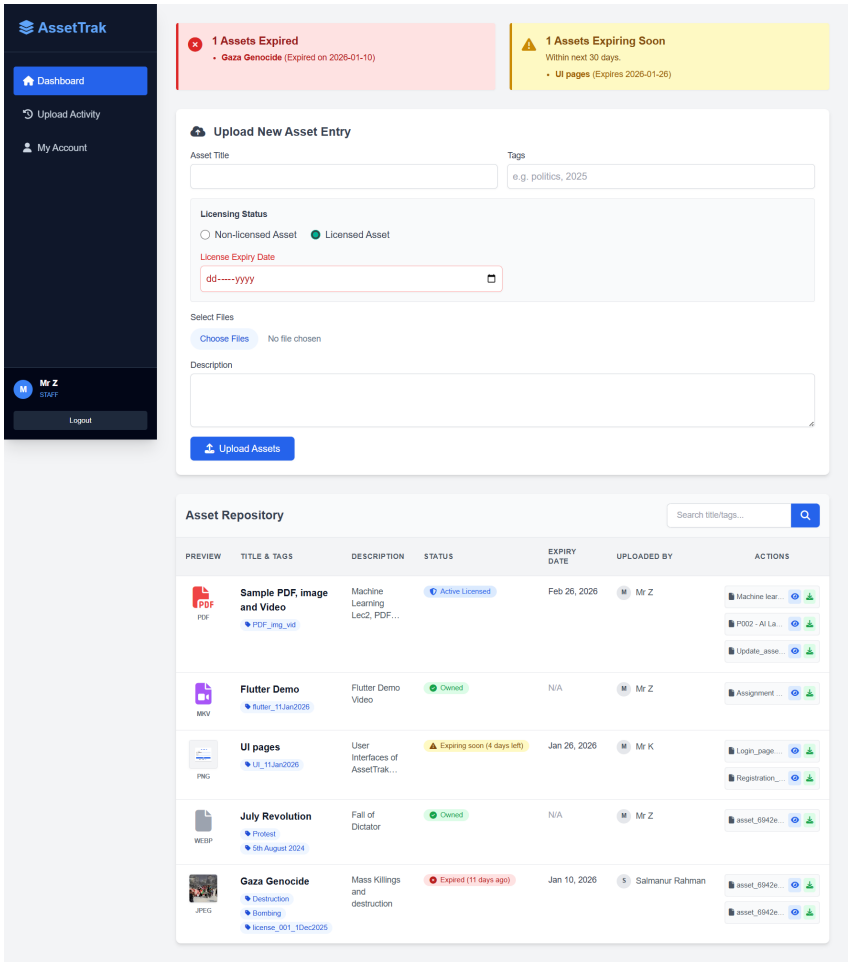


Figure 5.5.3.2 : Dashboard Page for Staff

5.5.4 My Account Page

Allows users to view profile details and **update passwords securely**. A profile card showing the user's role and a 'Change Password' form).

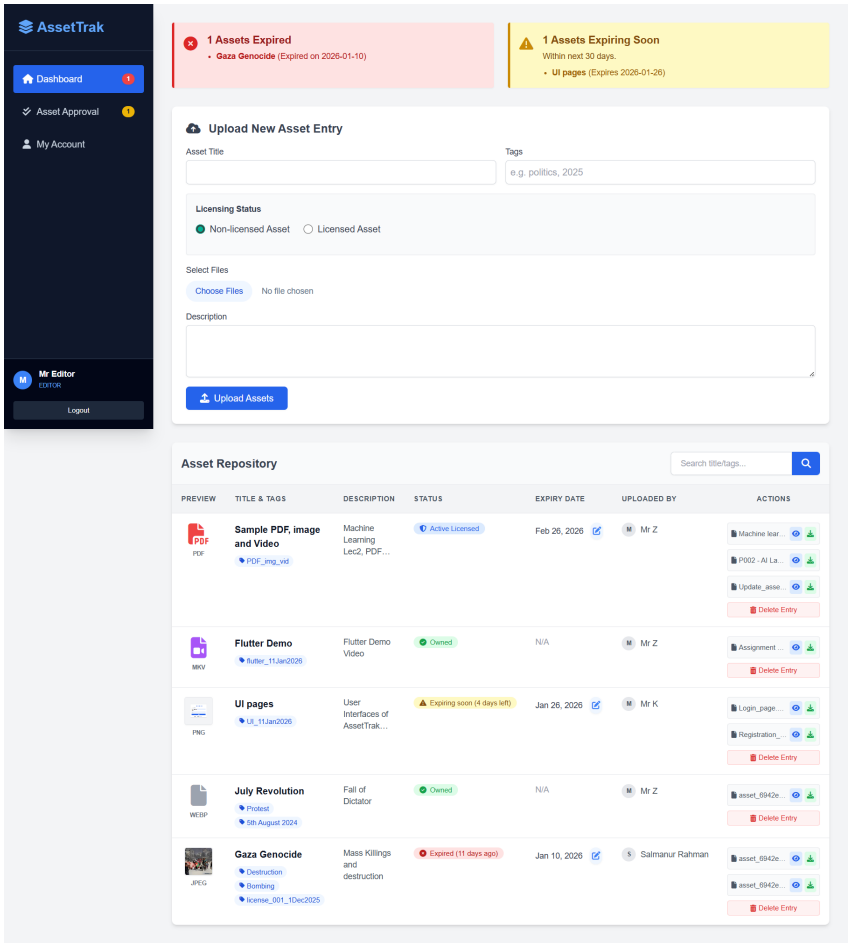


Figure 5.5.3.3 : Dashboard Page for Editor

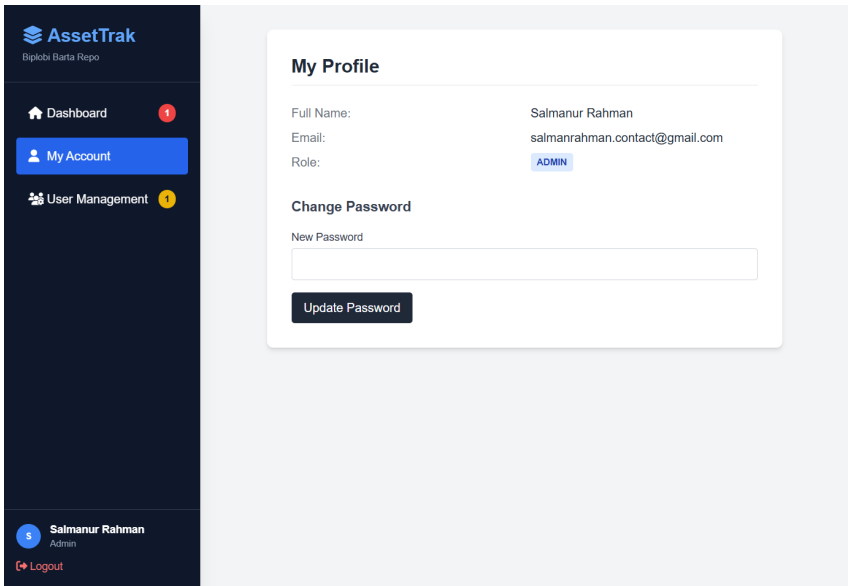


Figure 5.5.4 : My Account Page

5.5.5 User Approvals Management Page (Admin Only)

A dedicated interface for Admins to approve/reject new users with user role assignment. A table of approved users is showing on active accounts section with name, email, role and delete button to remove any user.

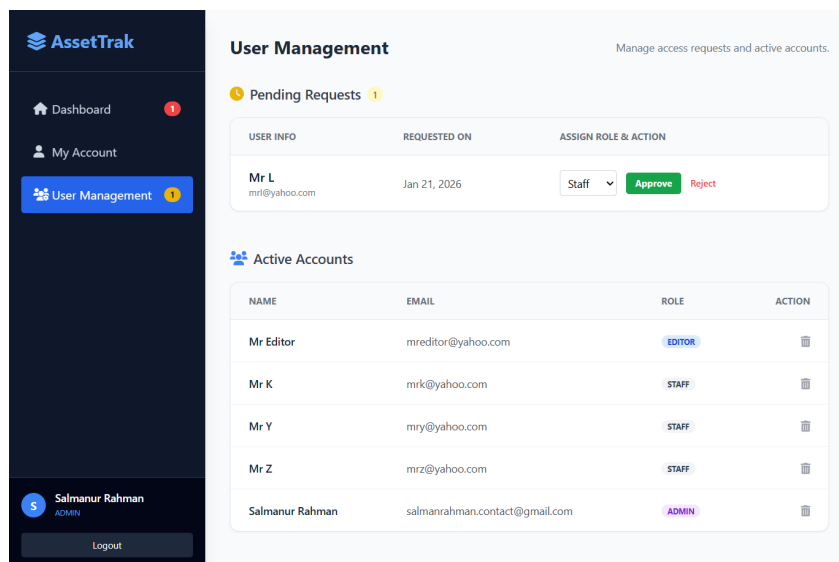


Figure 5.5.5 : User Approvals Management Page

5.5.6 Upload Activity Page (Staff Only)

A dedicated interface for Staffs to monitor their asset upload activity and status. Staff users can see their asset upload request status. Staff users will get rejection message if editor rejects an upload request.

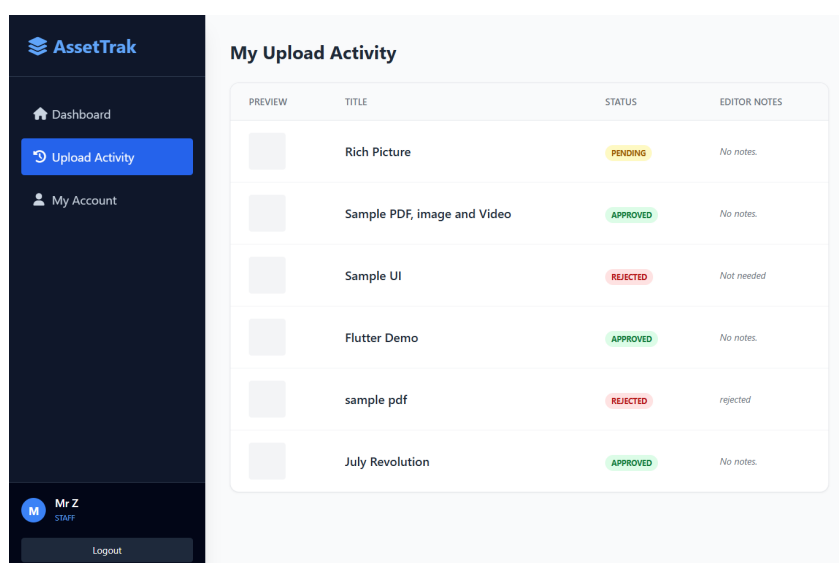


Figure 5.5.6 : Upload Activity Page

5.5.7 Asset Approvals Page (Editor Only)

A dedicated interface for editor to approve/reject asset upload request. A tableview showing asset details, username of uploader, upload timestamp and action section including view files button and confirm status button.

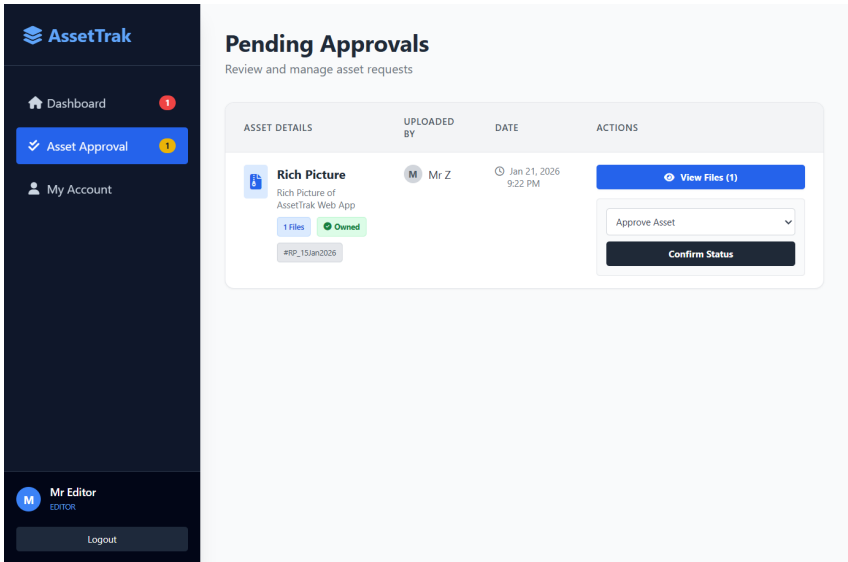


Figure 5.5.7 : Asset Approvals Page

5.6 Testing

5.6.1 Input and Output Analysis

Task Description	Input Data	Output / Result
User Login	Email & Password	Redirect to Dashboard & Session Start
Asset Upload	Title, Files, Expiry Date	Success Message & Table Update
Search Asset	Tag name (e.g., "Politics")	Filtered list of matching assets
Delete Asset	Click Delete + Confirmation	Asset & physical files removed

5.6.2 Designing Test Cases

Utilized **"Black Box Testing"** for UI outcomes and **"White Box Testing"** to verify database integrity after file deletions.

5.6.3 Test Results

ID	Description	Input	Expected Outcome	Status
TC-01	Unauthorized Access	Navigate to dashboard.php	Redirect to login.php	PASSED
TC-02	Expired Alert	Asset with past date	Red alert banner appears	PASSED
TC-03	File Integrity	Upload 3 images	3 download links in table	PASSED
TC-04	Role Security	Login as Staff	User Management hidden	PASSED

Chapter 6

Results and Analysis

In this chapter, we assess the application’s effectiveness and the technical benefits the **AssetTrak** project has provided for the editors at **Biplobi Barta**. By comparing the initial requirements against the final system performance, we measure the success of the transition from fragmented storage to a **centralized digital hub**.

6.1 Functional Outcomes

The primary achievement of the project was the successful transition of the core vision—from a conceptual design to a fully operational system. All **high-priority functional requirements** were met with high reliability:

- **Centralized Repository:** We successfully moved the organization away from physical hard drives. The system now acts as the **”Single Source of Truth,”** where any authorized user can access media from any browser within the office network.
- **The ”Safety Net” Alert System:** This was perhaps the most critical functional success. The logic comparing the current server date with the `license_expiry_date` works seamlessly. The dashboard now **automatically generates red banners** for expired content and yellow banners for assets nearing their 30-day limit, effectively removing the human error factor from legal compliance.
- **Batch Management:** The **”one-to-many”** database structure allows journalists to link multiple high-resolution files to a single descriptive title. This has simplified the upload process significantly, making it easy to keep raw footage and final edits grouped together.
- **Gatekeeper Security:** The **Admin approval system** ensures that the repository remains a closed ecosystem. Only staff vetted by the administrator can view or contribute to the database, maintaining high data integrity.

6.2 Performance Evaluation

In technical terms, **AssetTrak** was tested for its speed, responsiveness, and database performance:

- **Database Efficiency:** Through optimized **SQL queries** (which involved the use of `GROUP_CONCAT` to fetch more than one file path in one query pass), we managed to ensure that the dashboard loads in **less than a second** even with the increased number of assets. This has been tested in the local development environment using the **XAMPP stack**.
- **UI Fluidity:** The decision to use **Tailwind CSS** resulted in a very lightweight and responsive frontend. The interface feels snappy, and the **sidebar-based navigation** allows users to jump between the repository and their account settings without any noticeable lag.
- **System Stability:** During the testing phase, the **PHP backend** handled various file types (MP4, PNG, PDF) without errors. The **naming collision logic**—which renames files upon upload—proved effective in maintaining a clean and organized `/uploads/` directory on the server.

6.3 User Feedback

Reactions among the team at **Biplobi Barta** have been very encouraging. During a demonstration for the organizational supervisor, **Mr. MD Russel Hossain**, several key insights were gathered:

- **Ease of Discovery:** Journalists highlighted that the **"Global Search"** and tagging system are major time-savers. They mentioned that being able to type a keyword like "Archive" or "Interview" and get immediate results is a massive improvement over manually plugging in different USB drives.
- **Reduced Anxiety:** Editors specifically praised the **license alert system**. They expressed that having a visual warning for expiring content provides a sense of **"Peace of Mind,"** as they no longer have to maintain manual spreadsheets to track copyright dates.
- **Learning Curve:** The users appreciated how **user-friendly** the interface was, to the point where they did not require a manual to get started with it. The familiarity of the system, being similar to cloud storage solutions such as **Google Drive** or **Dropbox**, made it very easy for the employees.

6.4 Impact On Managing Digital Assets and Workflow

The implementation of **AssetTrak** has fundamentally shifted the operational culture at **Biplobi Barta** from a reactive model to a **proactive** one:

- **Workflow Synchronization:** In the past, the process was "siloed" in that files were locked away on personal computers. The process is now **synced** because an editor can see what the journalist has posted as soon as it reaches the server. This has eliminated the "**bottleneck**" of physical file transfers.
- **Legal Risk Mitigation:** The most profound impact is the **reduction of legal liability**. By centralizing license tracking, the organization has effectively closed the loophole that previously allowed for the accidental use of unlicensed media.
- **Archival Value:** The project has made it possible for a "messy" collection of documents to be scanned into a **searchable digital resource**. This means that rather than simply a means of storing documents, it is now a valuable resource for storing the organization's **media history**.

In conclusion, analysis has ensured that **AssetTrak** not only achieved its technological objectives but also ensured the delivery of a functional solution set to the **human organizational frustrations** explored at the commencement of the internship.

Chapter 7

Project as Engineering Problem Analysis

This chapter takes a step back to evaluate **AssetTrak** through the lens of **engineering responsibility**. Beyond just writing code, building a system for an organization like **Biplobi Barta** involves thinking about how the software survives over time, its impact on the people using it, and the **ethical guardrails** required to keep data safe and legal compliance intact.

7.1 Sustainability of the Project/Work

Sustainability in software engineering isn't just about the environment; it's about making sure the system doesn't become "**technical debt**" that nobody can fix in six months. **AssetTrak** has been created taking into consideration three pillars of sustainability:

- **Technical Scalability and Maintenance:** Firstly, a **modular approach** to the code makes the system maintainable and easier to manage as opposed to a monolithic form. For example, breaking the database connections into files such as `db.php` and the core functionalities into `functions.php` allows a future developer to modify the database details or security mechanisms without affecting the whole application.
- **Economic Viability:** By making the costs manageable for a local media company, the project utilizes a completely **open-source tech stack**. The system employs a combination of **PHP**, **MySQL** databases, and a front-end framework like **Tailwind CSS**.
- **Operational Longevity:** The system is **self-regulating** in terms of compliance. The "**Safety Net**" automated alarm system guarantees that the repository remains relevant from a legal perspective without needing constant maintenance from personnel. Since the code adheres to **web development best practices** and is compatible with **PHP 8.x**, it can scale effectively with the business.

7.2 Social and Environmental Effects and Analysis

Engineering solutions always ripple out into the real world. We analyzed how **AssetTrak** changes the office dynamic and its broader footprint:

- **Social Impact (Workflow and Mental Well-being):** Before this project, the "Old Way" was defined by high-pressure urgency and "fragmented storage." Journalists were stressed, digging through physical drives to meet deadlines. By providing a "Single Source of Truth," we've reduced that daily friction, fostering a more collaborative and organized workplace culture. It empowers staff to work faster while feeling **legally secure**.
- **Environmental Considerations:** Incorporating all virtual resources through a server solution such as **XAMPP** is far more **energy-efficient** compared to having numerous journalists employing multiple external hard drives and USB drives. Additionally, through optimizing database performance using **GROUP_CONCAT**, we ensure that servers use fewer **CPU cycles** over time, utilizing less energy through the organization's hardware resources.

7.3 Addressing Ethics and Ethical Issues

In the newsroom environment, the consideration of **ethics** is critical. These are the areas recognized where engineering considerations meet ethical responsibilities:

- **Data Privacy and Security:** The primary issue of data privacy relates to the staff's identity and login information. We have implemented **high-level encryption** through `password_hash()` so that in the event of a hack attempt on the database, no user can read the passwords. We also used **session-based access control** to ensure that sensitive media isn't accessible to anyone outside the vetted "Gatekeeper" system.
- **Intellectual Property and Copyright:** The exploitation of another's media source once the license has expired is an issue of **ethical incompetence** toward the original creator. The **License Expiry Alert** is designed to prevent editors from committing an injustice against a photographer or videographer by violating their **copyright**.
- **The "Gatekeeper" Responsibility:** The **Admin role** has immense authority in this context, especially in terms of approving or disapproving users. To avoid biased opinions and misuse, the system was developed to be **transparent**, displaying the status of all users to the Admin to make account management as professional as possible.

Chapter 8

Lesson Learned

Looking back on the **AssetTrak** development phase, the whole journey was as much about **professional growth and problem-solving** as about writing code. This chapter outlines the real-world hurdles encountered during the internship period at **Biplobi Barta** and how these were overcome.

8.1 Problems Faced During this Period

Developing a custom solution for a **fast-paced media environment** presented several unique challenges:

- **Mapping the "Chaos":** The first challenge was merely to comprehend the full extent of **decentralized storage** within the organization. From personal cloud accounts to labeled USB sticks, each journalist had their own way of saving files, making it hard to define a **standardized metadata schema** for all that would not be overly restrictive for them.
- **The "Large File" Bottleneck:** Due to the involvement of **high-definition video footage** and raw image files in this project, the initial struggle was with the local **XAMPP server environment**. The default PHP settings would often time out or throw errors during **batch uploads** of files larger than a few megabytes.
- **Database Structural Complexity:** The **"one-to-many"** relationship was a steep learning curve. I had to fathom how a user could upload five different files under one single **"Asset Title"** while making sure that if they searched for a tag, all the associated files would appear properly without **file search result duplication**.
- **Time Management vs. Feature Creep:** There were temptations throughout the 12-week internship to add **"extra" features** such as video previews or integrated editing tools. Balancing these desires against the core requirement of a **stable, secure repository** was a significant mental challenge.

8.2 Solution of those Problems

To move the project forward, I had to apply a mix of **technical research** and **collaborative communication**:

- **Visualization and Stakeholder Interviews:** To solve the requirement ambiguity, I relied on the **"Rich Picture" analysis**. By sitting down with the journalists and editors, I was able to map out exactly where the **"Legal Stress"** was coming from. This direct feedback helped me prioritize the **License Expiry Alert** over less critical cosmetic features.
- **Server-Side Configuration:** Regarding file upload issues, I was responsible for looking into **server configuration settings**. Setting `php.ini` configuration settings myself for image and video uploads helped ensure the system was optimized for **high-load sites**, such as in a newsroom configuration. Furthermore, I implemented **renaming code** to ensure that if a file with the same name was created by another author, it wouldn't overwrite the existing file.
- **SQL Mastery:** This one-to-many complexity was solved by using the `GROUP_CONCAT` function in **MySQL**. This allowed the system to fetch multiple file paths for a single metadata ID in a **single query**, which kept the dashboard fast and the code clean.
- **Mentorship and Iterative Testing:** I didn't solve these issues in isolation. Regular check-ins with my supervisor, **Mr. MD Russel Hossain**, and the use of **Black Box Testing** allowed me to catch bugs early. If a particular feature was not functioning as desired, I would **rollback** to the former modular version and enhance the logic while maintaining the **"Gatekeeper" security** feature throughout.

Finally, all these experiences have taught me that **engineering** is not just about perfection, but more about being **adaptable** and translating human needs into the **digital world**.

Chapter 9

Code

This chapter provides an overview of the technical foundation of **AssetTrak**. Instead of being a huge block of code, the system operates on a **module-based structure** where each script has a definite and distinct logic. This makes sure that the application developed will be **easily maintainable, secure, and scalable** to meet any requirement at **Biplobi Barta**.

9.1 Source Code

The following files represent the core **backend and frontend logic** of the application:

```
db.php > ...
1  <?php
2  // db.php Connects to MySQL and sets up constants.
3  session_start();
4
5  define('DB_HOST', 'localhost');
6  define('DB_USER', 'root');
7  define('DB_PASS', '');
8  define('DB_NAME', 'asset_trak_web_app_db');
9
10 // System Constants
11 define('MIN_PASS_LENGTH', 8);
12 define('EXPIRING_SOON_DAYS', 30);
13 define('ROLE_ADMIN', 'admin');
14 define('ROLE_STAFF', 'staff');
15 define('STATUS_PENDING', 'pending');
16 define('STATUS_APPROVED', 'approved');
17
18 $conn = new mysqli(DB_HOST, DB_USER, DB_PASS, DB_NAME);
19 if ($conn->connect_error) {
20     die("Connection failed: " . $conn->connect_error);
21 }
22
23 $upload_dir = 'uploads/';
24 >
```

Figure 9.1.1: db.php

db.php (Database Connection & Global Constants): This script is basically a **bridge file** through which the whole system operates. It sets up a connection between the **MySQL database** and encompasses global constants. This approach allows an individual to make adjustments to the server database details in **one file**, without having to revisit the whole code.

```

1 // functions.php
2 // functions.php Handles flash messages and basic auth checks.
3
4 function set_flash_message($type, $msg) {
5     $_SESSION['flash_message'] = ['type' => $type, 'text' => $msg];
6 }
7
8 function display_flash_message() {
9     if (isset($_SESSION['flash_message'])) {
10         $msg = $_SESSION['flash_message'];
11         unset($_SESSION['flash_message']);
12         $bgClass = $msg['type'] === 'success' ? 'bg-green-600' : 'bg-red-600';
13         $icon = $msg['type'] === 'success' ? 'fa-check-circle' : 'fa-exclamation-circle';
14
15         echo "
16         <div id='Flash-msg' class='fixed top-5 right-5 z-100 px-6 py-4 rounded shadow-lg text-white $bgClass'>
17             <div class='flex items-center'>
18                 <i class='fas $icon mr-3'></i>
19                 <span> . htmlspecialchars($msg['text']) . </span>
20             <button onclick='document.getElementById('Flash-msg').remove()'> class='ml-4 text-white hover:text-gray-200'><i class='fas fa-times'></i></button>
21         </div>
22         ";
23     }
24 }
25
26 function check_auth() {
27     if (isset($_SESSION['user_id']) || $_SESSION['user_status'] !== 'approved') {
28         header('Location: login.php');
29         exit;
30     }
31 }
32
33 function check_admin() {
34     if (isset($_SESSION['user_role']) || $_SESSION['user_role'] !== 'admin') {
35         header('Location: dashboard.php'); // Redirect non-admins back to dashboard
36         exit;
37     }
38 }

```

Figure 9.1.2: functions.php

functions.php (Helper Functions like Flash Messages): This script is the “utility belt” of this project. It houses global logic, including **security checks** like `check_auth` to protect pages and the **notification system** used to display success or error “flash messages” to the user.

```

1 // sidebar.php
2 // sidebar.php Contains the navigation and logic to calculate badge numbers automatically.
3
4 // Calculate counts for badges
5 $pending_user_count = 0;
6 $total_alert_count = 0;
7
8 if (isset($_SESSION['user_role']) && $_SESSION['user_role'] === 'admin') {
9     $count_stmt = $conn->query("SELECT COUNT(*) as count FROM users WHERE status = 'pending'");
10     $pending_user_count = $count_stmt->fetch_assoc()['count'];
11 }
12
13 $today_str = date('Y-m-d');
14 $license_str = date('Y-m-d', strtotime("+" . EXPIRING_500H_DAYS . " days"));
15 $alert_sql = "SELECT COUNT(*) as count FROM assets_meta WHERE is_licensed = 1 AND license_expiry_date IS NOT NULL AND (license_expiry_date < '$today_str' OR license_expiry_date > '$license_str')";
16 $alert_stmt = $conn->query($alert_sql);
17 $total_alert_count = $alert_stmt->fetch_assoc()['count'];
18
19 $current_page = basename($_SERVER['PHP_SELF']);
20
21 //
22
23 <aside class='w-64 bg-slate-900 text-white flex flex-col fixed h-full shadow-xl z-10'>
24     <div class='p-6 border-b border-slate-700'>
25         <h1 class='text-2xl font-bold text-blue-400'><i class='fas fa-layer-group'></i> AssetTrak</h1>
26         <p class='text-xs text-slate-400 mt-1'>©2024 Barto Repco</p>
27     </div>
28     <nav class='flex-1 p-4 space-y-2'>
29         <a href='dashboard.php' class='flex items-center justify-between px-4 py-3 rounded hover:bg-slate-800'><?php echo $current_page === 'dashboard.php' ? 'bg-blue-600' : ''> Dashboard</a>
30         <div class='flex items-center'>
31             <span class='bg-red-500 text-white text-xs font-bold px-2 py-1 rounded-full'><?php echo $total_alert_count; ></span>
32         </div>
33         <a href='my_account.php' class='flex items-center px-4 py-3 rounded hover:bg-slate-800'><?php echo $current_page === 'my_account.php' ? 'bg-blue-600' : ''> My Account
34         <a href='user_management.php' class='flex items-center justify-between px-4 py-3 rounded hover:bg-slate-800'><?php echo $current_page === 'user_management.php' ? 'bg-blue-600' : ''> User Management</a>
35     </nav>
36 </aside>

```

Figure 9.1.3: sidebar.php

sidebar.php (The Navigation Menu with Dynamic Badge Counters): To maintain a consistent look and feel across the app, this file **unifies the navigation**. It dynamically displays the **user’s role** and includes the logic for **session management**, ensuring the sidebar reflects the current user’s identity.

```

login.php <--
1 <?php
2 // login.php The default entry point.
3 require_once 'db.php';
4 require_once 'functions.php';
5
6 if (isset($_SESSION['user_id']) && $_SESSION['user_status'] === STATUS_APPROVED) {
7     header("Location: dashboard.php");
8     exit;
9 }
10
11 if ($_SERVER['REQUEST_METHOD'] === 'POST') {
12     $email = trim($_POST['email']);
13     $password = $_POST['password'];
14
15     $stmt = $conn->prepare("SELECT id, full_name, password_hash, role, status FROM users WHERE email = ?");
16     $stmt->bind_param("s", $email);
17     $stmt->execute();
18     $result = $stmt->get_result();
19
20     if ($user = $result->fetch_assoc()) {
21         if (password_verify($password, $user['password_hash'])) {
22             if ($user['status'] === STATUS_APPROVED) {
23                 $_SESSION['user_id'] = $user['id'];
24                 $_SESSION['user_name'] = $user['full_name'];
25                 $_SESSION['user_role'] = $user['role'];
26                 $_SESSION['user_status'] = $user['status'];
27                 header("Location: dashboard.php");
28                 exit;
29             } else {
30                 set_flash_message('error', 'Account pending approval.');
```

Figure 9.1.4: login.php

login.php (The Default Entry Page): This is the **first interaction point** for users. It handles **credential validation** against the database and specifically checks the **”Gatekeeper” status** to ensure that only **Admin-approved accounts** can start a session.

```

register.php <@html
1 <?php
2 // register.php
3 require_once 'db.php';
4 require_once 'functions.php';
5
6 if ($_SERVER['REQUEST_METHOD'] === 'POST') {
7     $full_name = trim($_POST['full_name']);
8     $email = trim($_POST['email']);
9     $password = $_POST['password'];
10    $confirm_password = $_POST['confirm_password'];
11
12    if ($password !== $confirm_password) {
13        set_flash_message('error', 'Passwords do not match.');
```

Figure 9.1.5: register.php

register.php (Registration Page): This page allows **new staff members** to request access. It captures user details and implements **security best practices** by using `password_hash()` to encrypt passwords before they are stored in the database with a **”pending” status**.

dashboard.php (Main Asset Repository Page): The **central hub** of the application where users interact with the media archive. It manages the **batch upload process**, displays

```

1 dashboard.php > ...
2 <?php
3 // dashboard.php The main app area. Includes Sidebar, Upload Logic, and Asset Table
4 require_once 'db.php';
5 require_once 'functions.php';
6 check_auth();
7
8 $is_admin = ($SESSION['user_role'] === ROLE_ADMIN);
9
10 // --- POST REQUESTS (Actions) ---
11 if ($SERVER['REQUEST_METHOD'] === 'POST') {
12
13     // UPLOAD
14     if (isset($_POST['action']) && $_POST['action'] === 'upload') {
15         $title = trim($_POST['title']);
16         $desc = trim($_POST['description']);
17         $tags = trim($_POST['tags']);
18         $is_licensed = isset($_POST['is_licensed']) ? (int)$_POST['is_licensed'] : 0;
19         $expiry_date = ($is_licensed === 1 && !empty($_POST['expiry_date'])) ? $_POST['expiry_date'] : null;
20
21         $conn->beginTransaction();
22         try {
23             $stmt = $conn->prepare("INSERT INTO assets.meta (user_id, title, description, tags, is_licensed, license_expiry_date) VALUES (?, ?, ?, ?, ?, ?)");
24             $stmt->bind_param("ississ", $SESSION['user_id'], $title, $desc, $tags, $is_licensed, $expiry_date);
25             $stmt->execute();
26             $asset_meta_id = $conn->insert_id;
27             $stmt->close();
28
29             if (!is_dir($upload_dir)) mkdir($upload_dir, 0777, true);
30             $file_count = count($_FILES['asset_files']['name']);
31             $uploaded_count = 0;
32             $stmt = $conn->prepare("INSERT INTO asset_files (asset_meta_id, file_path, file_type, original_name) VALUES (?, ?, ?, ?)");
33
34             for ($i = 0; $i < $file_count; $i++) {
35                 if ($_FILES['asset_files'][$i]['error'] !== 0) {
36                     $original_name = basename($_FILES['asset_files'][$i]['name']);
37                     $file_ext = strtolower(pathinfo($original_name, PATHINFO_EXTENSION));
38                     $new_filename = uniqid('asset_', true) . '.' . $file_ext;
39                     $target_path = $upload_dir . $new_filename;

```

Figure 9.1.6: dashboard.php

the searchable asset table, and renders the critical **License Expiry Alert** banners.

```

1 user_management.php > ...
2 <?php
3 // user_management.php Admin only page.
4 require_once 'db.php';
5 require_once 'functions.php';
6 check_auth();
7 check_admin();
8
9 // POST ACTIONS
10 if ($SERVER['REQUEST_METHOD'] === 'POST') {
11     if (isset($_POST['action']) && in_array($_POST['action'], ['approve_user', 'reject_user', 'delete_user'])) {
12         $target_user_id = (int)$_POST['user_id'];
13         if ($target_user_id === $SESSION['user_id']) {
14             set_flash_message('error', 'You cannot delete your own admin account.');

```

Figure 9.1.7: user management.php

user_management.php (Admin User Approval Page): A restricted interface reserved strictly for **Administrators**. It provides the tools to **approve or reject** new registrations and allows for the modification of user roles between **”Staff”** and **”Admin”**.

```

1 my_account.php > ...
2 <?php
3 // my_account.php
4 require_once 'db.php';
5 require_once 'functions.php';
6 check_auth();
7
8 // Fetch current user data for display
9 $stmt = $conn->prepare("SELECT * FROM users WHERE id = ?");
10 $stmt->bind_param("i", $SESSION['user_id']);
11 $current_user = $stmt->get_result()->fetch_assoc();
12
13 if ($SERVER['REQUEST_METHOD'] === 'POST' && isset($_POST['action']) && $_POST['action'] === 'update_profile') {
14     $new_pass = $_POST['new_password'];
15     if (!empty($new_pass) && strlen($new_pass) >= MIN_PASS_LENGTH) {
16         $hash = password_hash($new_pass, PASSWORD_DEFAULT);
17         $stmt = $conn->prepare("UPDATE users SET password_hash = ? WHERE id = ?");
18         $stmt->bind_param("si", $hash, $SESSION['user_id']);
19         $stmt->execute();
20         set_flash_message('success', 'Password updated successfully.');

```

Figure 9.1.8: my account.php

my_account.php (Profile & Password Page): This self-service page allows users to manage their own presence in the system. It displays their current role and provides a **secure** form to update their password.

```

1 // get_files.php
2 // get_files.php [Used by the "View Files" modal to fetch data asynchronously (if you plan to re-implement AJAX later) or you can rely on the data already loaded in the dashboard]
3 require_once 'db.php';
4 require_once 'functions.php';
5
6 if (!isset($_SESSION['user_id'])) {
7     echo json_encode([]);
8     exit;
9 }
10
11 if (isset($_GET['asset_id'])) {
12     header('Content-Type: application/json');
13     $aid = (int)$_GET['asset_id'];
14     $stmt = $conn->prepare("SELECT file_path, file_type, original_name FROM asset_files WHERE asset_meta_id = ?");
15     $stmt->bind_param("i", $aid);
16     $stmt->execute();
17     $res = $stmt->get_result();
18     $files = [];
19     while($row = $res->fetch_assoc()) { $files[] = $row; }
20     echo json_encode($files);
21 }
22
23

```

Figure 9.1.9: get_files.php

get_files.php (Backend Script for the View Files modal): This script handles the "child" side of the one-to-many relationship. It is called **asynchronously** to fetch and display the specific **file paths** and **download links** associated with a single asset metadata ID.

```

1 // upload_activity.php
2 // upload_activity.php [Used by the "Upload Activity" modal to fetch data asynchronously (if you plan to re-implement AJAX later) or you can rely on the data already loaded in the dashboard]
3 require_once 'db.php';
4 require_once 'functions.php';
5 check_auth();
6
7 $user_id = $_SESSION['user_id'];
8 $sql = "SELECT am.*,";
9         "SELECT CONCAT(file_path, '/', file_type, '/', original_name) FROM asset_files WHERE asset_meta_id = am.id LIMIT 1) as main_file";
10 FROM assets_meta am
11 WHERE am.user_id = $user_id
12 ORDER BY am.uploaded_at DESC;
13 $result = $conn->query($sql);
14
15 <DOCTYPE html>
16 <html lang="en">
17
18 <head>
19     <meta charset="UTF-8">
20     <title>My Upload Activity - AssetTrak</title>
21     <script src="https://cdn.tailwindcss.com"~></script>
22     <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.4.0/css/all.min.css">
23 </head>
24
25 <body class="bg-gray-50 flex">
26     <div class="flex-1">
27         <div class="text-2xl font-bold mb-6 text-gray-800">My Upload Activity</div>
28         <div class="bg-white rounded-xl shadow-sm border border-gray-200 overflow-hidden">
29             <table class="w-full text-left">
30                 <thead class="bg-gray-50 border-b">
31                     <tr>
32                         <th class="px-6 py-4 text-xs font-sans text-gray-500 uppercase">Preview</th>
33                         <th class="px-6 py-4 text-xs font-sans text-gray-500 uppercase">Title</th>
34                         <th class="px-6 py-4 text-xs font-sans text-gray-500 uppercase">Status</th>
35                         <th class="px-6 py-4 text-xs font-sans text-gray-500 uppercase">Editor Notes</th>
36                     </tr>
37                 </thead>
38                 <tbody>
39                     <tr>
40                         <td class="px-6 py-4">
41                         <div class="flex items-center">
42                             <div class="flex-shrink-0 w-100px h-100px">
43                                 <img alt="Thumbnail image of the uploaded file" data-bbox="42 42 100 100"/>
44                             <div class="flex flex-col justify-between p-2">
45                                 <div class="text-sm font-sans text-gray-500">Preview</div>
46                                 <div class="text-sm font-sans text-gray-500">Title</div>
47                                 <div class="text-sm font-sans text-gray-500">Status</div>
48                                 <div class="text-sm font-sans text-gray-500">Editor Notes</div>
49                             </div>
50                         </div>
51                     </td>
52                     <td class="px-6 py-4">
53                     <div class="flex items-center">
54                         <div class="flex-shrink-0 w-100px h-100px">
55                             <img alt="Thumbnail image of the uploaded file" data-bbox="54 54 112 112"/>
56                         <div class="flex flex-col justify-between p-2">
57                             <div class="text-sm font-sans text-gray-500">Preview</div>
58                             <div class="text-sm font-sans text-gray-500">Title</div>
59                             <div class="text-sm font-sans text-gray-500">Status</div>
60                             <div class="text-sm font-sans text-gray-500">Editor Notes</div>
61                         </div>
62                     </div>
63                 </tbody>
64             </table>
65         </div>
66     </div>
67 </body>
68

```

Figure 9.1.10: upload_activity.php

upload_activity.php (Upload History): A user-facing file that allows staff members to track the **status of their contributions**. It retrieves all records linked to the **current user's session ID**, displaying a list of uploads with visual status badges (**Pending, Approved, Rejected**). It also provides transparency by displaying specific **feedback or rejection reasons** entered by editor during the review process.

```

asset_approval.php > @hand
1  <?php
2  require_once 'db.php';
3  require_once 'functions.php';
4  check_auth();
5
6  // Only allow Editor and Admin
7  if ($SESSION['user_role'] != ROLE_EDITOR && $SESSION['user_role'] != ROLE_ADMIN) {
8      header("Location: dashboard.php");
9      exit;
10 }
11
12 // Handle Approval/Rejection
13 if ($SERVER['REQUEST_METHOD'] == 'POST' && isset($_POST['action']) && $_POST['action'] == 'process_approval') {
14     $asset_id = (int)$_POST['asset_id'];
15     $status = $_POST['status']; // approved or rejected
16     $reason = isset($_POST['rejection_reason']) ? trim($_POST['rejection_reason']) : '';
17
18     $stmt = $conn->prepare("UPDATE assets_meta SET approval_status = ?, rejection_reason = ? WHERE id = ?");
19     $stmt->bind_param("ssi", $status, $reason, $asset_id);
20
21     if ($stmt->execute()) {
22         set_flash_message('success', 'Asset status updated successfully.');

```

Figure 9.1.11: asset_approval.php

asset_approval.php (Editor's Verification): A critical administrative file restricted to **Editor**. It queries the database for all **pending assets**, grouping associated files (PDFs, videos, images) for efficient review. It features a **preview modal** to verify content safety and quality, and provides the logic to **approve or reject** submissions, updating the **database status** and storing rejection reasons if necessary.

```

logout.php
1  <?php
2  // logout.php
3  session_start();
4  session_destroy();
5  header("Location: login.php");
6  exit;
7  ?>

```

Figure 9.1.12: logout.php

logout.php (Logout Logic): A straightforward but **essential security file**. It ends the user session, deletes all **credentials information** stored, and then **redirects the user** to the login page. It deletes all information of the user stored in his/her **session**, then redirects the user to the login page.

Chapter 10

Sustainability, Future Work & Conclusion

The final chapter will examine the bigger impact that the **AssetTrak** project has had, not only in the functionality of the code, but in the way that it will live on into the future.

10.1 Sustainability along with Environmental and Social Effects

The true success of an engineering project lies in its ability to remain useful and relevant without causing negative ripples. **AssetTrak** was developed with a focus on **three-dimensional impact**:

- **System Sustainability:** Firstly, we have been able to make this project **economically and technically sustainable** by using a **PHP architecture** that is based on modules and using **open source technology**. This is because it will be free of licensing costs, hence it will be sustained forever. Its **standard-compliant code** means that a new developer can step in and understand the logic in hours, preventing the system from becoming **”technical debt.”**
- **Environmental Considerations:** Centralizing assets on a single, **optimized server environment** is a much **”greener”** approach than the previous model. In the **”Old Way,”** dozens of journalists were running multiple external hard drives and high-powered personal machines to sift through data. By moving to a **centralized hub** and utilizing **optimized SQL queries** (like `GROUP_CONCAT`), we’ve lowered the overall **energy consumption** and hardware strain required to manage the organization’s media.
- **Social Impact:** The **social transformation** at **Biplobi Barta** has been the most visible outcome. We have replaced a culture of **”deadline panic”** and **”search frustration”** with one of **professionalism and security**. The **”Safety Net”** license alerts have provided the editors with **”Peace of Mind,”** knowing they are no longer at risk of

accidental copyright infringement. This reduction in workplace stress has created a more **collaborative and focused newsroom environment**.

10.2 Future Works

Though **AssetTrak** is a strong system as of now with all functionalities intact, there are a number of possible improvements it can offer:

- **Cloud Infrastructure Integration:** The next step would be to move from the local **XAMPP environment** to a **cloud-based setup** (for instance, **AWS S3** or **Azure**). This would provide the benefit of **global access**, where the journalist would be able to upload footage from the field, while the editor could assess the resources at home.
- **AI-Driven Auto-Tagging:** Our plan is to incorporate **Machine Learning APIs** that have the capability of automatically scanning uploaded pictures and videos to produce **descriptive tags**. This would make the **"Global Search"** feature more effective by not requiring manual entry of data.
- **Mobile Companion App:** A mobile app could enable journalists to record content in **high definition** on their mobile device and upload it directly to the metadata of the repository in **real time**.
- **Advanced Media Processing:** Future revisions may integrate the **video transcoding facility** for the purpose of producing a **"proxy" view**, enabling the user to view the video in the browser before downloading the full-sized image.

10.3 Conclusion

The development of **AssetTrak** was a journey that began with a **"Rich Picture"** of organizational chaos and ended with a **"Single Source of Truth."** Throughout the internship at **Biplobi Barta**, there has been a continued focus on filling the gap that exists between **technical competence and human need**.

The project has met all the **essential requirements**, right from the secure **"Gatekeeper"** login functionality to the **one-to-many upload feature**. More importantly, it solved the critical problem of **license tracking**, ensuring that the organization is **legally protected and operationally efficient**.

In conclusion, **AssetTrak** is more than just a **Digital Asset Management tool**; it is a solution that respects the rights of creators, protects the interests of the media house, and **empowers journalists** to focus on storytelling rather than file searching. It stands as a successful example of how **modular web development** can resolve deep-seated organizational frustrations and pave the way for a more digital, organized future.

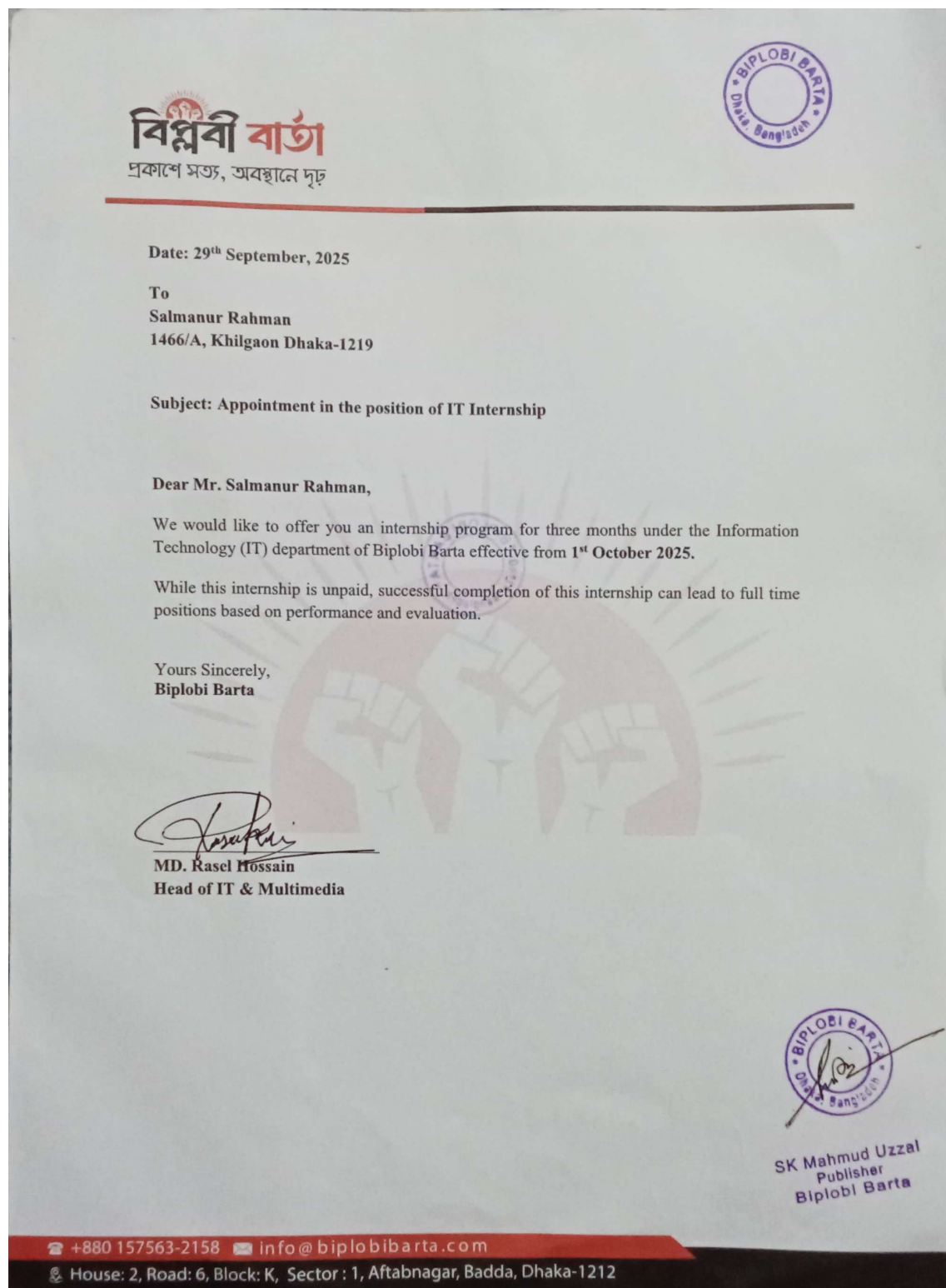


Figure 10.3.1: Internship Offer Letter

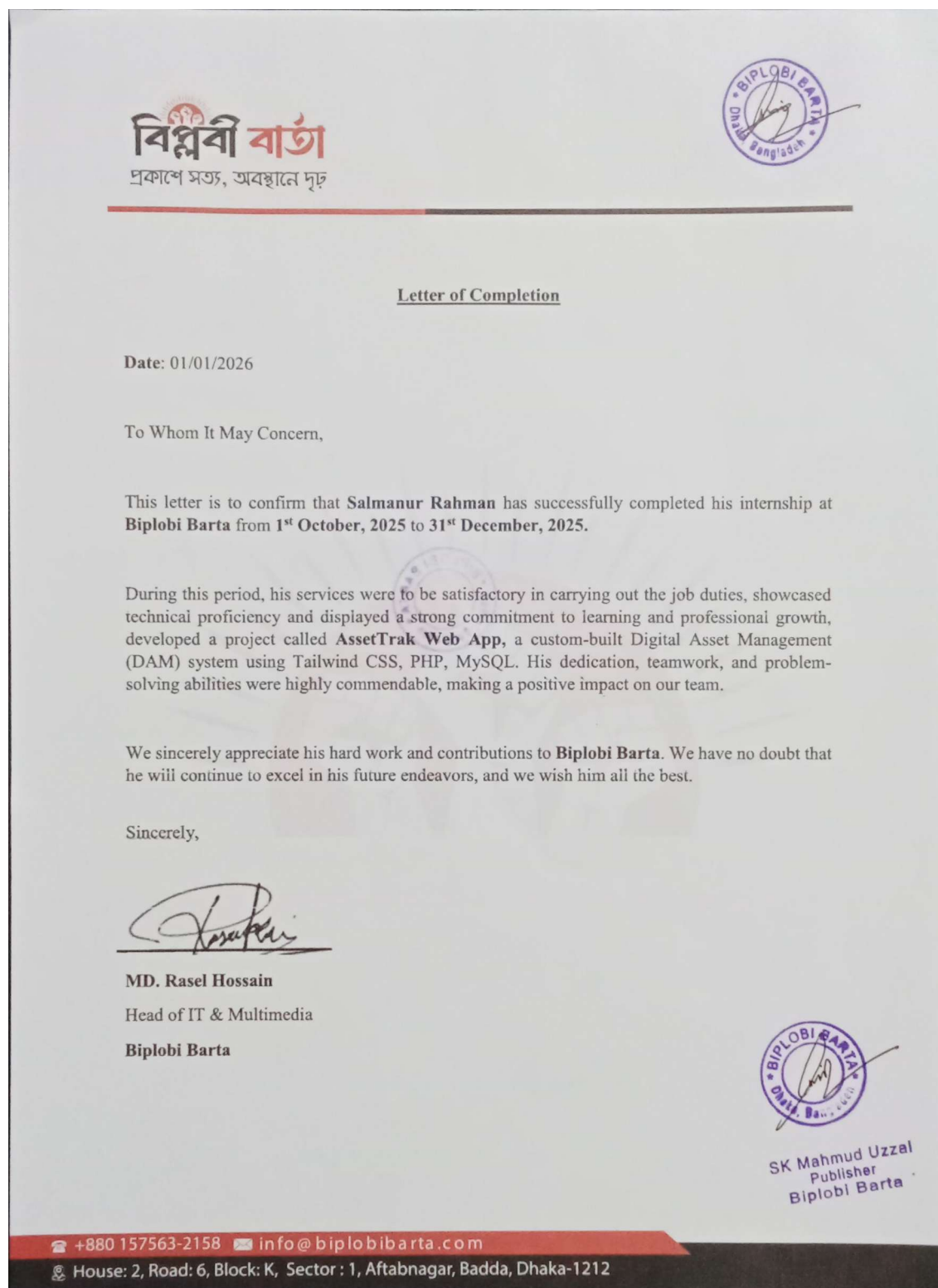


Figure 10.3.2: Letter of Completion

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Salmanur Rahman

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By

Salmanur Rahman

Student ID: **2021806**

Autumn, 2025

The student modified the internship final report as per the recommendation made by his or her academic supervisor and/or panel members during final viva, and the department can use this version for achieving.

Signature of the Supervisor

Bijoy Rahman Arif

Lecturer A

Department of Computer Science & Engineering

School of Engineering, Technology & Sciences

Independent University, Bangladesh



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Supervisor:

Bijoy Rahman Arif

Lecturer A

Department of Computer Science & Engineering
Independent University, Bangladesh

January 14, 2026

Dissertation submitted in partial fulfillment for the degree of Bachelor of
Science in Computer Science

Department of Computer Science & Engineering
Independent University, Bangladesh

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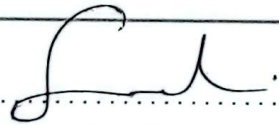
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Department of Computer Science & Engineering

School of Engineering, Technology & Sciences

Independent University, Bangladesh